DuoGesture: Neuro-Inspired and Biomechanically Informed Dual-Stream Co-Speech Gesture Generation

Anonymous Author(s)
Affiliation
Address
email

Abstract

Co-speech gesture generation requires both semantic expressivity and biomechanically plausible rhythmic motion. Existing holistic gesture models mix lexically grounded semantic gestures with frequent prosody-aligned beat gestures. This limits semantic grounding, speech-motion alignment, and kinematic smoothness. We propose *DuoGesture*, a neuro-inspired and biomechanically informed dual-stream approach that decomposes co-speech gesture synthesis into coupled semantic and beat streams. The two streams are coordinated by a *Semantic Variational Information Bottleneck*, a stochastic frame-level gate that learns when semantic gestures should override rhythmic beat motion. The semantic stream is controlled by *Motion-Grounded Semantic Conditioning*, which replaces purely linguistic word embeddings with motion-language representations to provide motion-aligned semantic priors for long-tailed lexical triggers of gestures. The beat stream is further regularised by an *Inertial Beat Prior*, an anthropometry-weighted arm-chain module that reduces jitter and improves rhythmic consistency without constraining semantic frames. Objective evaluations and subjective experiments show that DuoGesture outperforms strong holistic baselines, while component ablations confirm the complementary roles of semantic grounding, stochastic stream selection, and biomechanical regularisation. Demos and qualitative illustrations are available on the anonymous project page: https://duogesture.github.io/DuoGesture/.

1 Introduction

Speech and co-speech gestures form an integrated communicative system, yet gesture types play different roles and are processed accordingly. Reliable co-speech gesture generation is therefore important for embodied agents, virtual avatars, and accessible speech-driven animation systems, where gestures affect perceived naturalness, communicative clarity, and user trust. Beat gestures align with prosody and rhythm, supporting turn-taking and the flow of interaction, while semantic gestures (deictic, iconic, metaphoric) are related to lexical content and occur more sparsely [McNeill, 1992]. Cognitive neuroscience further indicates that manual gesture comprehension recruits partially dissociable but interacting networks: parietal–premotor circuits associated with visuomotor structure, and inferior-frontal/temporal circuits implicated in communicative meaning and gesture–speech integration [Tipper et al., 2015, Arbib, 2012]. We adopt this dual-process view as a computational stance, where co-speech gesture should be modelled as the interaction of two coupled streams rather than as a single motion process. This stance is different from most current holistic generators, which treat gestures as a single homogeneous stream [Zhang et al., 2025, Liu et al., 2025a], resulting in *three persistent limitations*: (i) misalignment between gesture and speech timing, (ii) poor semantic expressivity, and (iii) jittery beat motion that lacks biomechanical principles.

We argue that these limitations stem from three modelling drawbacks. First, current architectures rely on a single mechanism to model all gesture frames Liu et al. [2025a, 2024a, 2022, 2025b], even

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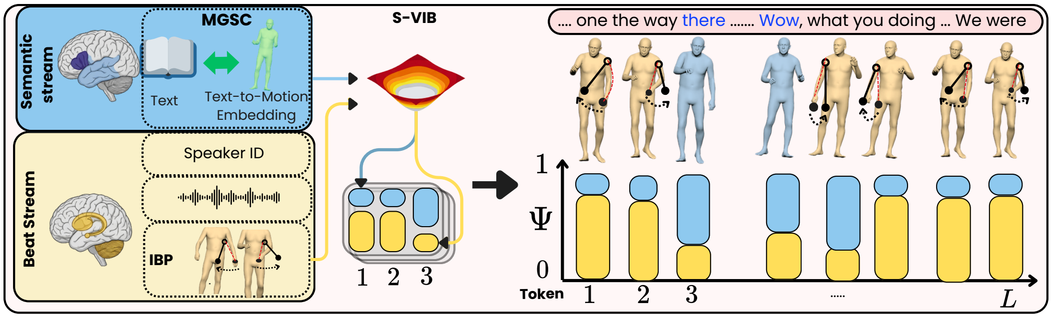

Figure 1: DuoGesture models co-speech gestures as coupled *semantic* and *beat* streams coordinated by a stochastic frame-level weight. It integrates *motion-grounded semantic conditioning* and an *inertial prior* to ensure semantic expressivity and rhythmic smoothness.

though beat and semantic gestures are distinct. Second, semantic conditioning is usually derived from linguistic embeddings, which encode *what was said* rather than *what motion should accompany it*. This is especially problematic for long-tailed semantic triggers that appear rarely, or never, during training. Third, beat gestures are learned without physical or biomechanical constraints. These choices create gaps: a *semantic gap*, where text embeddings provide weak priors for what a gesture should look like, and a *beat gap*, where rhythmically aligned gestures may achieve reasonable average objective metrics but remain prone to over-smoothing and acceleration artefacts.

To overcome these limitations and gaps, we propose *DuoGesture*, a dual-stream model shown in Figure 1, that separates lexically grounded semantic motion from prosody-aligned beat motion while allowing them to interact. Supported by our auxiliary analysis in Sec. A, which shows distinct temporal and kinematic profiles for semantic- and beat gestures, DuoGesture uses stream-specific conditioning and regularisation rather than a single homogeneous generator. The semantic stream is conditioned by *Motion-Grounded Semantic Conditioning* (MGSC), which leverages a pretrained text-to-motion representation to provide motion-aligned cues for long-tailed lexical triggers of gestures. The beat stream is regularised by an *Inertial Beat Prior* (IBP), an anthropometry-weighted arm-chain loss that encourages smooth rhythmic motion while remaining inactive on semantic frames. The streams are coupled by a *Semantic Variational Information Bottleneck* (S-VIB), which learns when semantic motion should override beat motion. Our novel contributions are summarised as follows:

- We formulate a dual co-speech gesture generation model that models lexically grounded semantic motion and prosody-aligned beat motion as coupled but distinct processes.
- We introduce **MGSC**, a motion-grounded semantic conditioning module that uses pretrained text-to-motion representations to improve semantic cues for sparse lexical gesture triggers.
- We design **IBP**, an anthropometry-weighted arm-chain regulariser that reduces beat-motion jitter and improves rhythmic consistency without constraining semantic frames.
- We introduce **S-VIB**, a stochastic frame-level gating mechanism that learns when to activate the semantic stream while avoiding deterministic always-on gate collapse.
- We show on BEAT2 that DuoGesture improves distributional realism, as measured by Fréchet Gesture Distance (FGD), in both single- and multi-speaker settings, while maintaining a competitive trade-off across beat alignment, diversity, and facial-motion metrics.

2 Related Work

Holistic Co-speech Gesture Generation. Current work has been shaped by two-stage hierarchical quantisation paradigms. Early architectures, e.g., by TalkSHOW [Yi et al., 2023] and ProbTalk [Liu et al., 2024b], established VAE-based stochastic priors and category-contingent decoding for multifaceted body dynamics. This trajectory toward granularity was extended by EMAGE [Liu et al., 2024a], which proposed a spatially-decoupled tokenisation scheme across facial, manual, and corporal streams. SOTA benchmarks are currently defined by PyraMotion [Yin et al., 2025], which leverages a multi-resolution Anchor-based Pyramid VQ-VAE to minimise distributional divergence, yielding a competitive FGD. However, these extant models largely operate under the assumption of *kinematic homogeneity*, treating gesticulation as a singular stochastic process and thereby blurring the functional divergence between rhythmic prosody and semantic morphology. While SemTalk [Zhang et al., 2025] introduced a first-order approximation of this divergence via frame-level soft gating, its efficacy remains bottlenecked by shallow linguistic conditioning and a lack of physical regularisation within its homogeneous motion stream.

Cross-Modal Semantic Grounding Gesture Generation. A major gap in co-speech gesture generation is the *linguistic-kinematic gap*, where idiosyncratic word embeddings (e.g., BERT [Devlin et al., 2019]) are insufficient to encapsulate the morphology of human motion. To solve this, Semantic Gesticulator [Zhang et al., 2024] leverages LLM-based lexical parsing, yet the resulting representations remain inherently decoupled from the physical motion manifold. While GestureDiffuCLIP [Ao et al., 2023] employs CLIP latents to enforce semantic consistency, it operates within a text-centric latent space, often overlooking the fine-grained temporal dynamics of gesticulation. The RAG-Gesture [Mughal et al., 2025] introduced non-parametric synthesis by retrieving explicit motion exemplars at inference time. However, lookups in high-dimensional spaces incur significant computational latency and memory usage. Our work resolves this; instead of performing an explicit search, we *distil* the structured knowledge of a motion-aligned encoder (TM [Petrovich et al., 2023]) into our generative pipeline. By anchoring semantics directly within a kinematic manifold, **DuoGesture** achieves the representational richness of retrieval-based methods while maintaining the inference efficiency of purely parametric frameworks, particularly for long-tail lexical distributions.

Physically-Consistent Motion Synthesis. While physics-based constraints have been explored in character animation via differentiable simulators or post-hoc projections [Peng et al., 2018, Yuan et al., 2023], their integration as inductive biases within gesture models remains under-explored. In this work, we diverge from computationally expensive simulation-based approaches by incorporating *anthropometric priors* directly into the generator's rhythmic stream. Specifically, we leverage De De Leva's segment mass distributions [De Leva, 1996] to derive joint-specific exponential moving average time constants. This formulation provides a computationally efficient, training-time-only regularisation that imposes no inference overhead. Crucially, our empirical validation (Sec. 3.2.3) confirms biomechanical literature suggesting that conversational gesticulation is primarily governed by the *arm-chain-spine* pendulum dynamics.

3 Method

DuoGesture is a two-stage latent generator. Stage 1 (Sec. 3.1) is a regional RVQ-VAE tokeniser. Stage 2 is the contribution of this paper: a dual-stream generator with a stochastic frame-level weight, motivated by the fact that semantic and beat gestures are statistically and physically distinct, and a single shared network cannot optimise them simultaneously. We address this with three contributions: *(i) Motion-Grounded Semantic Conditioning (MGSC)* (Sec. 3.2.1) grounds semantic conditioning in text-to-motion latent space, decoupling it from the long-tailed word distribution; *(ii) Semantic Variational Information Bottleneck (S-VIB)* (Sec. 3.2.2) predicts the per-frame weights Ψ via a variational bottleneck, preventing gate collapse; and *(iii) Inertial Beat Prior (IBP)* (Sec. 3.2.3) is a training-time velocity-consistency regulariser on the proximal arm chain that injects an anthropometric inductive bias into the beat stream.

3.1 Problem Formulation

A motion sequence of length L is split into four body regions $\mathcal{R} = \{\mathrm{hand}, \mathrm{upper}, \mathrm{lower}, \mathrm{face}\}$, $\mathbf{G} = \{\mathbf{G}^r\}_{r\in\mathcal{R}}$ with $\mathbf{G}^r \in \mathbb{R}^{L\times J_r}$. Stage 2 conditions on HuBERT audio features $e_a \in \mathbb{R}^{L\times 1024}$, a speaker identity embedding ID, motion-grounded semantic features $\mathbf{S}^m \in \mathbb{R}^{L\times 256}$ the per-frame output of the MGSC module (Sec. 3.2.1) and a 4-frame seed pose $\tilde{\mathbf{p}}$. The target is the discrete latent code $\mathcal{Z}^q = \{\mathbf{Z}^q_r\}_{r\in\mathcal{R}}$ produced by the Stage 1 quantiser on the ground-truth motion.

Stage 1: Regional RVQ-VAE Tokeniser. We adopt the regional RVQ-VAE of Liu et al. [2024a], Zhang et al. [2025], Liu et al. [2025a, 2026]: one encoder–quantiser–decoder triple $(\mathcal{E}^r, \mathcal{Q}^r, \mathcal{D}^r)$ per region, trained with the standard reconstruction and codebook commitment losses. The stage-1 components are kept frozen once moving to Stage 2, decoupling (Stage 1) from temporal motion

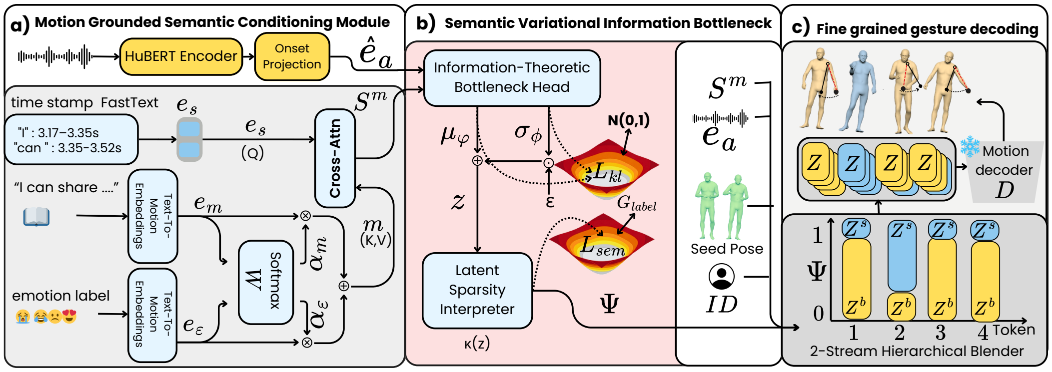

Figure 2: DuoGesture pipeline. (a) MGSC fuses lexical semantics, motion-style, and emotion embeddings through cross-attention to produce the motion-grounded semantic representation $\mathbf{S}^m$. (b) S-VIB combines $\mathbf{S}^m$ with the HuBERT timing projection $\hat{e}_a$ to infer when semantic gestures should be activated and what semantic content they should express. The bottleneck samples $\mathbf{z}\tau \sim q\phi(\mathbf{z}\tau)$ under $\mathcal{L}\mathrm{kl}$ regularisation, and maps it to the semantic gate Ψ with semantic supervision. (c) Fine-grained decoding blends beat codebooks Z_r^{b} (yellow) and semantic codebooks Z_r^{s} (blue) using Ψ.

synthesis (Stage 2), and enabling the generator to operate in a structured, low-dimensional latent space.

3.2 DuoGesture

DuoGesture separates beat and semantic motion and adaptively fuses them via a dedicated module. The *beat backbone* f_b produces per-region latents from audio, speaker identity, and seed pose alone:

$$f_b : \big(e_a,\ \mathrm{ID},\ \tilde{\mathbf{p}};\ \theta_b\big)\ \longrightarrow\ Z_r^{\mathrm{b}},\quad r \in \mathcal{R}, \tag{1}$$

where Z_r^{b} is the beat codebook per region. This beat branch is modulated by a semantic branch via a dedicated weighting module, namely *S-VIB*. First, the *MGSC* module assembles per-frame lexical (e_s), motion-style (e_m), and emotion (e_ε) embeddings into a motion-grounded semantic feature, and *S-VIB* produces per-frame the following:

$$f_{s-vib} : \big(e_s,\ e_m,\ e_\varepsilon,\ e_a;\ \theta_{s-vib}\big)\ \longrightarrow\ \big(\mathbf{S}^m,\ \Psi\big),\quad \mathbf{S}^m \in \mathbb{R}^{L\times 256},\ \Psi \in [0,1]. \tag{2}$$

We then use the semantics features $\mathbf{S}^m$ to impose semantics on beat frames' output through the semantic branch:

$$f_s : \big(e_a,\ \mathrm{ID}, \Psi,\ \mathbf{S}^m;\ \theta_s\big)\ \longrightarrow\ Z_r^{\mathrm{s}},\quad r \in \mathcal{R}; \tag{3}$$

where the output of the semantic branch Z_r^{s} is then injected into Z_r^{b} via a dedicated module to produce the fused codebook $Z_r = f_{fusion}\big(Z_r^{\mathrm{b}},\ Z_r^{\mathrm{s}}, \Psi; \theta_{fusion}\big)$. The resulting codebook is then fed into the Stage-1 decoder $\mathcal{D}^r$, which maps the result to joint space: $\hat{\mathbf{G}}^r = \mathcal{D}^r\big(Z_r\big)$.

3.2.1 Motion-Grounded Semantic Conditioning (MGSC)

To bridge the gap between abstract semantic representations and their corresponding kinetic realisations, we propose the MGSC module. Existing gesture-generation methods typically use off-the-shelf text or vision-language encoders, for example FastText Bojanowski et al. [2017], BERT Devlin et al. [2019], or CLIP Radford et al. [2021], whose representations are learned from linguistic or image-text supervision rather than from body-motion dynamics. In contrast, MGSC introduces a motion-grounded conditioning pathway that aligns semantic cues with their kinetic realisations. As illustrated in Fig. 2(a), MGSC produces a per-frame semantic feature $\mathbf{S}^m \in \mathbb{R}^{L\times 256}$ that conditions the semantic generator g_{sem} and feeds S-VIB. It assembles three streams, all projected to 256 dimensions: (i) $e_s \in \mathbb{R}^{256}$, a per-frame FastText Bojanowski et al. [2017] embedding of the word spoken at frame l, from BEAT2's forced-alignment timestamps; (ii) e_m, an utterance-level motion-style embedding from Text-To-Motion Petrovich et al. [2023]; and (iii) e_ε, an emotion embedding from

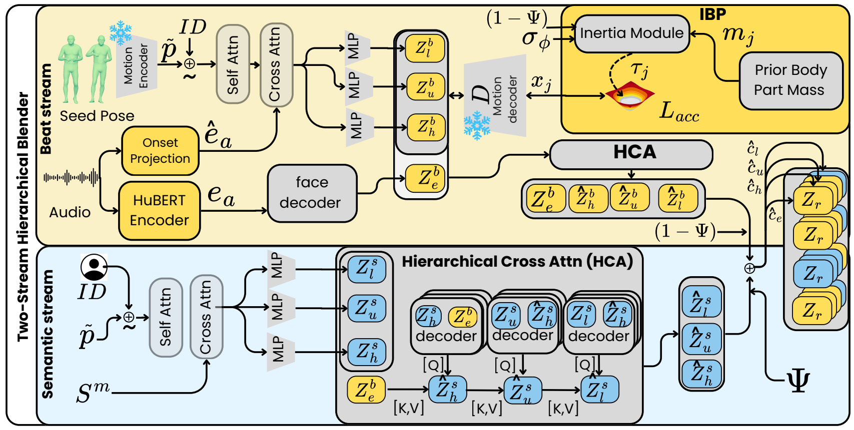

Figure 3: Two-Stream Hierarchical Blender. The beat stream encodes the seed pose $\tilde{\mathbf{p}}$ and speaker ID to predict region-wise beat latents, with the face handled by a separate audio-conditioned decoder. Arm-chain beat latents are decoded during training and regularised by IBP through a τ_j-weighted smoothness loss. The semantic stream conditions parallel region-wise latents on the gated semantic feature. Hierarchical cross-attention refines both streams across body regions.

Text-To-Motion, using emotion labels provided by BEAT2 Liu et al. [2022]. e_m and e_ε are blended by a learned softmax gate into a fused memory $\mathbf{m}$:

$$\boldsymbol{\alpha} = \mathrm{softmax}\left(\mathbf{W}_\alpha[e_m; e_\varepsilon]\right), \quad \mathbf{m} = \alpha_{(m)} e_m + \alpha_{(\varepsilon)} e_\varepsilon. \tag{4}$$

e_s then queries this memory via cross-attention to produce the final semantic feature: $\mathbf{S}^m = \mathrm{MLP}\left(\mathrm{CrossAttn}\left(Q = e_s,\ K = V = \mathbf{m}\right)\right) \in \mathbb{R}^{L \times 256}$.

3.2.2 Semantic Variational Information Bottleneck (S-VIB)

We want a gate to select a semantic gesture at the right moment. For this, the gate needs to consider what and when to gesture semantically. At the same time, this bottleneck (gate) should not collapse into a single path: either all semantic or all beat. For this reason, we propose S-VIB. As illustrated in Fig. 2(b), S-VIB operates on two streams: the MGSC output $\mathbf{S}_m \in \mathbb{R}^{256}$ (*what* to gesture, derived from e_s) and a low-capacity HuBERT timing projection $\hat{e}_a \in \mathbb{R}^{64}$ (*when* to gesture). $\hat{e}_a$ is obtained by passing HuBERT features e_a through a convolutional encoder and bottlenecking to 64 dimensions; it is entirely independent of the FastText word embedding e_s.

The **Information-Theoretic Bottleneck Head** maps $\mathbf{S}^m$ and $\hat{e}_a$ to two 16-dimensional outputs via separate linear heads: a mean $\boldsymbol{\mu}_\phi \in \mathbb{R}^{16}$ and a log-variance $\log \boldsymbol{\sigma}_\phi^2 \in \mathbb{R}^{16}$. A stochastic sample is drawn via the reparameterisation trick, $\mathbf{z} = \boldsymbol{\mu}_\phi + \exp\left(\frac{1}{2}\log \boldsymbol{\sigma}_\phi^2\right) \odot \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$, and passed to the **Latent Sparsity Interpreter** κ (two-layer MLP, $\kappa : \mathbb{R}^{16} \to \mathbb{R}^2$), which outputs two-dimensional beat/semantic logits; the semantic probability is the per-frame gate: $\Psi = \mathrm{softmax}\left(\kappa(\mathbf{z})\right)_{\mathrm{sem}} \in [0, 1]$. The outputs $\boldsymbol{\mu}_\phi$ and $\log \boldsymbol{\sigma}_\phi^2$ are regularised against the standard Gaussian prior $\mathcal{N}(0, I)$ via a KL divergence Alemi et al. [2017], Kingma et al. [2016]:

$$\mathcal{L}_{kl} = D_{\mathrm{KL}}\left(\mathcal{N}(\boldsymbol{\mu}_\phi, \boldsymbol{\sigma}_\phi^2) \parallel \mathcal{N}(0, I)\right) = \frac{1}{2}\sum_{d=1}^{Z}\left(\mu_{\phi,d}^2 + e^{\log \sigma_{\phi,d}^2} - \log \sigma_{\phi,d}^2 - 1\right). \tag{5}$$

In practice, we apply a per-dimension free-bits floor Kingma et al. [2016] ($\lambda_{\mathrm{fb}} = 0.50$ nats, $Z = 16$), whose per-dim KL falls below λ_{fb} receive no gradient, preventing the optimiser from over-regularising uninformative dimensions while preserving capacity for those that carry semantic signal. The gate Ψ is trained to predict the per-frame semantic annotations $s_\tau \in \{0, 1\}$ via a semantic loss $\mathcal{L}_{sem}$.

3.2.3 Two-Stream Hierarchical Blender

The Two-Stream Hierarchical Blender (Fig. 3) produces per-region beat residuals Z_r^{b} and semantic residuals Z_r^{s}, decoded by the frozen Stage-1 decoders. The two streams are structurally parallel but differ in their conditioning and regularisation, reflecting the asymmetry between beat and semantic gesture production established in Sec. A. **Beat stream**, (top, yellow in Fig. 3) takes as input the masked seed pose $\tilde{\mathbf{p}}$ embedding $\in \mathbb{R}^{T \times 768}$, where we add speaker identity ID and periodic positional encodings are applied before a self-attention pass. Speech onset projection ($\hat{e}_a$) is integrated with the resulting embeddings through a cross-attention layer. Three region-specific MLPs then project the refined beat region latents Z_u^{b}, Z_l^{b}, Z_h^{b}. The face latent Z_e^{b} is obtained from a separate face decoder conditioned on the HuBERT audio feature e_a.

Inertial Beat Prior (IBP). On beat stream, we apply our proposal to constrain the upper-body and hand movements with biomechanical constraints. For this reason, the beat latents Z_u^{b}, Z_l^{b}, Z_h^{b} are decoded by the frozen Stage-1 decoder into raw (rot6d) poses $\mathbf{x}_{j,t}$ and an **Inertia Module** (fed by body-part masses m_j, gate $(1-\Psi)$, and VIB posterior variance σ_ϕ^2) produces per-joint weights τ_j that scale $\mathcal{L}_{acc}$. IBP is a training-time regulariser that teaches the beat-stream network θ_b to produce latents that decode into physically smooth poses. IBP penalises deviation from constant-velocity motion. The loss is the mean squared error against the constant-velocity prediction $\hat{\mathbf{x}}_{j,t} = 2\mathbf{x}_{j,t-1} - \mathbf{x}_{j,t-2}$, weighted by $\tau_{j,t}$: $\mathcal{L}_{acc} = \mathbb{E}_{j,t}\left[\tau_{j,t} \left\|\mathbf{x}_{j,t} - \hat{\mathbf{x}}_{j,t}\right\|^2\right]$. Rather than penalising all joints equally, IBP weights each joint by a smoothing coefficient derived from De Leva's anthropometric masses De Leva [1996] and the S-VIB gate:

$$\tau_{j,t} = \tau_{\mathrm{base}} \cdot \sqrt{\frac{m_j}{m_{\mathrm{max}}}} \cdot (1 - \Psi_t) \cdot (1 + \alpha\, \sigma_{\phi,t}^2), \tag{6}$$

where m_j is the De Leva body-segment mass fraction for joint j, m_{max} is the fraction of the heaviest segment (spine1/abdomen, ≈ 0.163 of body mass), so $\sqrt{m_j/m_{\mathrm{max}}} \in (0, 1]$ is a sqrt-compressed relative inertial weight; $\tau_{\mathrm{base}} = 0.5$ is the maximum smoothing applied to the heaviest arm-chain joint on a pure beat frame. Hence, training with this regulariser encourages smoother latents without any explicit physics at inference time. The face latent Z_e^{b} does not pass through the IBP block since facial motion has distinct dynamics that should not be constrained by IBP.

Semantic stream (bottom, blue in Fig. 3) operates on MGSC output $\mathbf{S}^m$. We first add masked seed pose embeddings ($\tilde{\mathbf{p}}$) and ID, along with periodic positional encodings, before a self-attention pass. The resulting embeddings are then integrated with $\mathbf{S}^m$ through cross attention. Then three region-specific MLPs project these into semantic region latents Z_u^{s}, Z_h^{s}, Z_l^{s}.

Inspired by Zhang et al. [2025], a *Hierarchical Cross-Attention (HCA)* block refines these latents by allowing each region to attend to its sibling semantic latents: the hands decoder attends to $Z_u^{\mathrm{s}} + Z_l^{\mathrm{s}}$ as `[K,V]`, the upper decoder attends to $Z_h^{\mathrm{s}} + Z_l^{\mathrm{s}}$, and the lower decoder attends to $Z_u^{\mathrm{s}} + Z_h^{\mathrm{s}}$, producing final semantic latents $\hat{Z}_u^{\mathrm{s}}$, $\hat{Z}_h^{\mathrm{s}}$, $\hat{Z}_l^{\mathrm{s}}$. HCA is applied similarly on the beat codebooks. Finally, as illustrated in Fig. 2(c), for each of the three body regions $r \in \{h, u, l\}$, the beat residual Z_r^{b} (yellow token, $\Psi \approx 0$) and the semantic residual Z_r^{s} (blue token, $\Psi \approx 1$) are blended frame-by-frame via a fusion function. The fused and face latents are then quantised by nearest-neighbour lookup in the RVQ codebook:

$$Z_r = (1 - \Psi)\hat{Z}_r^b + \Psi \hat{Z}_r^s, \qquad \hat{c}_{r,t} = \arg \min_{k \in \{1,\ldots,K\}} \left\|e_k - Z_{r,t}\right\|_2^2,$$

where e_k denotes the k-th codebook vector and $\hat{c}_{r,t}$ is the selected discrete token for region r at frame t. The selected tokens are decoded by the frozen Stage-1 decoder as $\hat{G}_r = D_r(\hat{c}_r)$. Finally, our **full training objective** is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{lat}} + \mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{sem}} + \beta_{\mathrm{vib}}\, \mathcal{L}_{\mathrm{kl}} + \beta_{\mathrm{phys}}\, \mathcal{L}_{\mathrm{acc}}, \tag{7}$$

where: $\mathcal{L}_{\mathrm{lat}}$ is the MSE between the predicted continuous latents and the Stage-1 VQ targets for all body regions; $\mathcal{L}_{\mathrm{cls}}$ is the cross-entropy over the four RVQ codebook levels for each region; $\mathcal{L}_{\mathrm{sem}}$ (Sec. 3.2.2) is the S-VIB gate supervised against BEAT2 semantic flags; $\mathcal{L}_{\mathrm{kl}}$ (Sec. 3.2.2) is the VIB bottleneck KL with a free-bits floor; and $\mathcal{L}_{\mathrm{acc}}$ (Sec. 3.2.3) is the IBP inertia residual, active only on beat frames via $\tau_{j,t}$. The scalars β_{vib} and β_{phys} are warmup-scheduled weights.

4 Experiments

Dataset. We evaluate DuoGesture on BEAT2 [Liu et al., 2024a], a standard benchmark for co-speech gesture generation. It contains approximately 76 hours of speech, motion, facial expression, and speaker identity annotations from 30 speakers, covering diverse expressive behaviours. Unlike other datasets, such as TalkSHOW [Yi et al., 2023], BEAT2 provides frame-level annotations of gesture type (including semantic and beat gestures) and eight emotion categories, making it suitable for our study. Following the standard BEAT2 protocol used in prior holistic gesture-generation work [Liu et al., 2024a, Mughal et al., 2025], we report results in two settings: *(i) a single-speaker setting on Speaker 2* and *(ii) a multi-speaker setting over 25 speakers*. Unless otherwise stated, the data are split into training, validation, and test partitions at 85%/7.5%/7.5%. The **implementation details** are elaborated in Sections B and C of the Appendices.

Table 1: Overall comparison on BEAT2 (single-speaker, top; all-speaker, bottom). FGD is the primary metric; BA, Diversity, MSE, and LVD are secondary diagnostics. The table illustrates that high BA or Diversity alone does not imply better perceived gesture quality: several methods obtain strong secondary scores but have substantially worse FGD.

Setting	Model	FGD $\times 10^{-1}\downarrow$	BA $\times 10^{-1}\uparrow$	Diversity $\uparrow$	MSE $\times 10^{-3}\downarrow$	LVD $\times 10^{-5}\downarrow$
One Speaker	DiffStyleGesture [Yang et al., 2023] (IJCAI 2023)	8.866	7.239	11.13	–	–
	AMUSE [Chhatre et al., 2024] (CVPR 2024)	12.11	**8.318**	**14.93**	–	–
	SynTalker [Chen et al., 2024a] (ACM MM 2024)	5.366	7.812	13.05	–	–
	HoloGest [Cheng and Huang, 2025] (3DV 2025)	5.341	7.957	14.15	–	–
	RAG-Gesture [Mughal et al., 2025] (CVPR 2025)	8.08	7.34	11.97	–	–
	Habibie et al. [Habibie et al., 2021] (IVA 2021)	9.040	7.716	8.213	8.614	8.043
	DiffSHEG [Chen et al., 2024b] (CVPR 2024)	8.986	7.142	11.91	7.665	8.673
	ProbTalk [Liu et al., 2024b] (CVPR 2024)	5.040	7.711	13.27	8.617	–
	MambaTalk [Xu et al., 2024] (NeurIPS 2024)	5.366	7.812	13.95	**6.289**	**6.897**
	SemTalk [Zhang et al., 2025] (ICCV 2025)	4.278	7.770	12.91	7.153	6.938
	PyraMotion [Yin et al., 2025] (NeurIPS 2025)	4.612	7.420	13.42	7.176	7.270
	Ours (DuoGesture)	**4.101**	7.557	12.34	7.103	7.646
All Speakers	TalkSHOW [Yi et al., 2023] (CVPR 2023)	6.145	6.863	**13.12**	7.791	7.771
	GestureLSM [Liu et al., 2025b] (ICCV 2025)	4.268	5.250	11.20	–	–
	EMAGE [Liu et al., 2024a] (CVPR 2024)	5.643	**7.707**	12.92	7.694	7.593
	SemTalk [Zhang et al., 2025] (ICCV 2025)	5.214	7.689	12.74	7.612	**7.498**
	Ours (DuoGesture)	**4.081**	7.699	12.83	**7.502**	7.658

Evaluation Metrics. We evaluate holistic co-speech gestures along four axes: distributional realism, speech–motion synchrony, motion variation, and facial stability. We treat user studies and Fréchet Gesture Distance (FGD) [Yoon et al., 2020] as the primary evaluations, since FGD compares generated and real motion distributions in a learned gesture-feature space, and is the standard metric with the strongest reported perceptual support, where FGD was the only objective metric found to correlate with subjective human-likeness ratings [Kucherenko* et al., 2024].

We report Beat Alignment (BA) [Li et al., 2021b], pairwise L1 Diversity [Li et al., 2021a], facial MSE, and L1 Vertex Difference (LVD) [Xing et al., 2023] as secondary diagnostic metrics. These metrics are informative but not sufficient indicators of perceptual quality. High BA can reflect exaggerated beat-like motion, and high Diversity can reflect large motion variance rather than plausible or semantically appropriate gestures. Accordingly, we use a ***Pareto criterion***: a preferable model should reduce FGD without achieving this gain at the expense of severe degradation in other metrics.

Comparison Methods. We benchmark DuoGesture against a broad set of representative holistic co-speech gesture generators, covering early speech-to-gesture models, hierarchical and diffusion-based methods, semantic-based approaches, and recent BEAT2 state-of-the-art systems. The comparison includes DiffuseStyleGesture [Yang et al., 2023], AMUSE [Chhatre et al., 2024], SynTalker [Chen et al., 2024a], HoloGest [Cheng and Huang, 2025], RAG-Gesture [Mughal et al., 2025], Habibie et al. [Habibie et al., 2021], DiffSHEG [Chen et al., 2024b], ProbTalk [Liu et al., 2024b], MambaTalk [Xu et al., 2024], TalkSHOW [Yi et al., 2023], EMAGE [Liu et al., 2024a], SemTalk [Zhang et al., 2025], GestureLSM [Liu et al., 2025b], and PyraMotion [Yin et al., 2025].

4.1 Quantitative Results

Overall comparison. Table 1 reports the BEAT2 comparison under the single-speaker and all-speaker protocols. We treat FGD as the primary realism metric and the remaining columns as secondary diagnostics; a preferable model should reduce FGD without severely degrading the others. In the single-speaker setting, DuoGesture achieves the lowest FGD (4.101), improving over SemTalk (4.278) and PyraMotion (4.612). It also improves MSE over PyraMotion (7.103 vs. 7.176) and is competitive on BA (7.557, between SemTalk's 7.770 and PyraMotion's 7.420). The trade-off

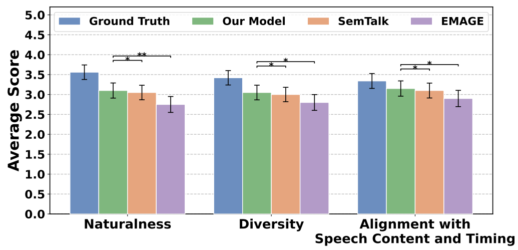

Figure 4: User study results comparing Ground Truth, DuoGesture, SemTalk, and EMAGE. Stars show significant differences.

Variant	MGSC	S-VIB	IBP	FGD ↓	BA ↑	Diversity ↑
(a) w/o MGSC (S-VIB + IBP only)	–	✓	✓	4.803	7.531	12.61
(b) MGSC only (linear σ-gate)	✓	–	–	4.306	7.551	12.52
(c) MGSC + S-VIB (no IBP)	✓	✓	–	4.178	7.446	12.77
(d) MGSC + IBP (linear σ-gate)	✓	–	✓	4.137	7.557	12.65
(e) Full DuoGesture	✓	✓	✓	**4.081**	**7.699**	**12.83**

Table 2: Component-wise ablation of DuoGesture on BEAT2 in the all-speaker setting.

appears on Diversity and LVD, where DuoGesture gives up a small margin to the strongest baselines. Methods that report substantially higher BA or Diversity (e.g., AMUSE, HoloGest) do so at much higher FGD, placing them at a different operating point rather than dominating DuoGesture. The all-speaker setting follows the same pattern. DuoGesture again reaches the best FGD, with the closest competitor, GestureLSM. DuoGesture's BA is effectively tied with EMAGE's (7.699 vs. 7.707), and it achieves the lowest MSE among the reported methods. Diversity (12.83) is slightly below EMAGE and above SemTalk, and LVD (7.658) is slightly above SemTalk's and EMAGE's; we view these as small concessions against the substantial FGD reduction.

Across both protocols, DuoGesture sits at the most realism-favourable point of the Pareto trade-off: it leads FGD by a clear margin while staying within a small tolerance of the best secondary scores. Methods that surpass it on individual secondary metrics do so at noticeably worse FGD, which prior work has consistently linked to perceptual quality [Kucherenko* et al., 2024].

Ablation Study. Table 2 isolates the contribution of each module on BEAT2 (all-speaker setting): each row removes or replaces one component while holding the others fixed. *MGSC* dominates the FGD gain: replacing motion-grounded semantic representation with standard semantic representations raises FGD from 4.081 to 4.803, an effect larger than removing S-VIB ($+0.056$) or IBP ($+0.097$). This is consistent with our hypothesis that motion-grounded semantic conditioning closes the linguistic–kinematic gap left open by purely lexical embeddings. *IBP* drives beat alignment. E.g., comparing variants (c) and (e), adding IBP on top of MGSC and S-VIB raises BA from 7.446 to 7.699, the largest BA delta in the table, confirming that the inertial prior contributes specifically to rhythmic regularity rather than to distributional realism. *S-VIB* protects diversity: replacing it with the deterministic σ-gate (variants b and d) lowers Diversity to 12.52 and 12.65 versus 12.83 for the full model, matching the design motivation that the variational bottleneck prevents the gate from collapsing to an always-on state and keeps the per-frame motion distribution wider. The ablation, therefore, supports the central design claim that semantic grounding, stochastic stream selection, and biomechanical beat regularisation are individually useful and most effective when combined.

4.2 Subjective and Qualitative Results

User study. We conduct a controlled perceptual study using 35-second clips from the BEAT2 test set. The study includes six narrated topics and compares Ground Truth, and representative state-of-the-art methods, namely, EMAGE, SemTalk, and DuoGesture. Thirty native English-speaking participants from the UK, balanced by self-reported gender (male:female = 1:1; mean age 38.6), evaluated 24 randomly ordered videos using a five-point Likert scale. Participants rated each video along three axes: naturalness, motion diversity, and alignment with speech content and timing. The order of methods and clips was randomised for each participant. As shown in Fig. 4, Ground Truth receives the highest scores, as expected. Among the generated visualisations, DuoGesture obtains the strongest overall perceptual ratings, with higher judgments than SemTalk and EMAGE across the evaluated axes. This result is consistent with the quantitative analysis: DuoGesture gives a better perceptual trade-off between realism, synchrony, and expressiveness. The user study, therefore, provides independent evidence that the Pareto-optimal objective profile observed in Table 1 corresponds to motions that users judge as more natural and better aligned with speech. Further details about the user study are provided in Section D of the Appendix.

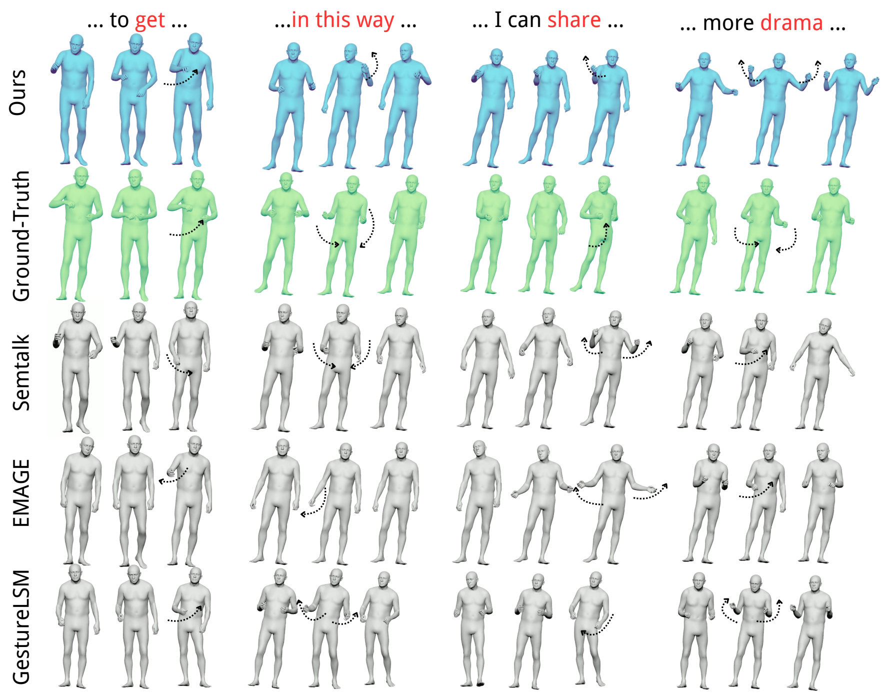

Figure 5: Qualitative comparison of co-speech gesture generation across semantic and beat-dominant speech contexts. We visualise motion sequences conditioned on representative phrases, including “to get”, “in this way”, “I can share”, and “more drama”.

4.3 Qualitative Results

Qualitative comparison. Figure 5 compares generated motion sequences for speech segments containing semantic and beat-dominant phrases. For semantic phrases such as “to get”, “I can share”, and “more drama”, DuoGesture produces gestures with clearer arm trajectories and more visible phrase-dependent structure than the compared baselines. SemTalk and GestureLSM tend to generate weaker or more ambiguous movements, whereas EMAGE often produces plausible but less semantically differentiated gestures. For the beat-dominant phrase “in this way”, the generated motion remains temporally coherent with the speech rhythm while avoiding excessive smoothing. These examples are consistent with the quantitative trend: DuoGesture improves semantic expressiveness and distributional realism, while maintaining competitive beat synchronisation.

5 Conclusion, Limitations, and Future Work

We presented *DuoGesture*, a neuro-inspired and biomechanically informed dual-stream framework for co-speech gesture generation. DuoGesture decomposes generation into a semantic stream for lexically grounded gestures and a beat stream for prosody-aligned rhythmic motion, with a stochastic frame-level gate coordinating their interaction. By combining motion-grounded semantic conditioning, stochastic stream selection, and an inertial beat prior, DuoGesture improves distributional realism on BEAT2 while preserving competitive alignment, diversity, and facial-motion fidelity. Ablations and user studies confirm that the three components provide complementary gains, supporting explicit semantic-beat decomposition as an effective design for co-speech gesture generation.

DuoGesture’s generalisation to other languages, cultures, speakers, recording conditions, and interaction settings has not been tested due to a lack of dataset availability in the co-speech gesture generation domain. MGSC depends on the coverage and biases of a pretrained text-to-motion representation, while IBP uses a biomechanical prior that may not capture full-body gestures, object interaction, or contact-rich motion. Future work should evaluate cross-dataset and multilingual generalisation, develop metrics that better capture communicative meaning, and extend the biomechanical prior beyond arm-chain beat motion.

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A Motion Analysis: Beat vs. Semantic Motion on BEAT2

Setup and controlled sampling. We analyse BEAT2 [Liu et al., 2022, 2024a] test-split motion (≥ 15-frame windows, all 25 speakers) using the dataset's per-frame semantic annotation flags to segment sequences into contiguous *beat windows* (flag $= 0$) and *semantic windows* (flag $= 1$). To ensure a fair, balanced comparison, we draw a matched sample: for each speaker we randomly draw the same number of beat and semantic windows (capped at the smaller set), yielding $\approx 1{,}870$ windows per class from 265 clips. All metrics are duration-weighted and accompanied by 95% CIs from hierarchical bootstrap (speaker $\rightarrow$ clip $\rightarrow$ window). The matched windows are representative of typical gesture production for each class (short rhythmic repetitions for beat, and clear lexical stroke events for semantic), and the class balance removes any confound from the natural 12% semantic sparsity.

Results. The two classes show clearly distinct kinematics (Table 3 and Fig. 6). Beat motion peaks at 1.12 Hz with moderate inter-joint coupling (PLV $= 0.31$), indicating that shoulder and forearm oscillate at partially independent rhythms, consistent with a damped-pendulum model. A constant-velocity (inertial) oscillator explains $R^2 = 0.41$ of beat variance, and this improves by $\Delta R^2 = +0.84$ when the mass grouping covers arm + core joints, but *degrades* by $\Delta R^2 = -0.95$ when all 55 joints are included, directly motivating IBP's arm-chain-only mask (Sec. 3.2.3). Semantic motion peaks at 1.69 Hz with strongly higher inter-joint coupling (PLV $= 0.53$, ΔPLV $= +0.22$), indicating holistic co-activation of the arm during a lexical stroke, a pattern better captured by shape-conditioned generation than by a mass-weighted smoother. Arm-swing PSD (Fig. 6) further shows that beat spectra have a narrow, peaked structure (half-BW 0.46 Hz, prominence $4.6\times$) while semantic spectra are $1.9\times$ broader and less tonal, confirming that beat motion is rhythmically regular and semantic motion is not.

B Technical Implementation

Data. BEAT2 [Liu et al., 2024a], 85%/7.5%/7.5% split, 64-frame clips at 30 fps (stride 20), 4-frame seed pose.

Training. Adam (lr $= 10^{-4}$, no weight decay), step LR decay ($\gamma = 0.3$), 200 epochs. Stage 2 trained on 4 GPUs A100 via PyTorch DDP, effective batch size 256.

Table 3: Kinematic statistics for matched beat and semantic windows (BEAT2 test split, 25 speakers; hierarchical bootstrap 95% CI in brackets).

Metric	Beat	Semantic
Shoulder peak (Hz)	1.12 [1.02, 1.23]	1.69 [1.56, 1.86]
PLV (shoulder $\leftrightarrow$ forearm)	0.308 [0.268, 0.349]	0.525 [0.487, 0.557]
Oscillator R^2	0.406 [0.388, 0.427]	0.509 [0.493, 0.524]
ΔR^2: arm+core vs. shoulder	$+0.84$	—
ΔR^2: all joints vs. shoulder	-0.95	—
Arm-swing half-BW (Hz)	0.46	0.89

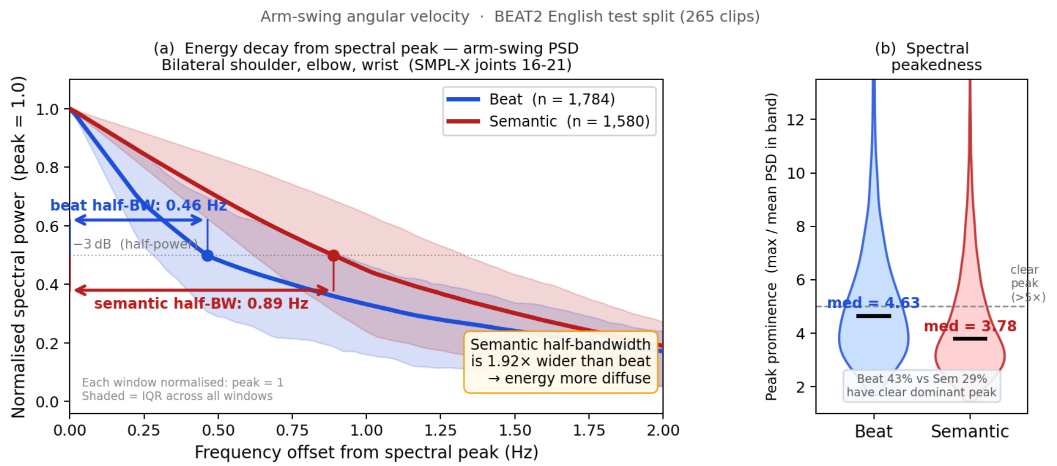

Figure 6: Arm-swing spectral analysis for bilateral shoulder/elbow/wrist joints (SMPL-X joints 16–21), matched beat and semantic windows (BEAT2 test split, $\approx$1,870 windows each, 25 speakers). **(a)** Mean normalised PSD decay from the spectral peak (shaded = IQR): beat spectra (blue) have a half-bandwidth of 0.46 Hz; semantic spectra (red) are $1.9\times$ broader (0.89 Hz), indicating diffuse rather than tonal energy. **(b)** Peak prominence (max/mean PSD in band): 43% of beat windows have a clear dominant peak ($> 5\times$) versus only 29% of semantic windows, confirming that rhythmic regularity is a beat-specific property.

Representations. Motion: 55 SMPL-X joints in rot6d (330-dim). Audio: HuBERT (1024-dim). Text: FastText [Bojanowski et al., 2017] (300-dim). All streams projected to 256-dim.

Stage 1. Regional RVQ-VAE [Liu et al., 2024a], codebook $C = 256$, 4 RVQ levels, frozen during Stage 2.

Stage 2. Transformer backbone: hidden size 768, 1 self-attention layer; 3 decoder layers and 1 codebook head per region. MGSC embeddings (e_m, e_ε) from TM [Petrovich et al., 2023] (256-dim). S-VIB: bottleneck $Z = 16$, timing projection 64-dim, KL warmup epochs 20–100 ($\beta_{\text{target}} = 0.01$), free-bits $\lambda_{\text{fb}} = 0.5$ nats, semantic boost $\rho = 3.0$. IBP: physics warmup epochs 30–80 ($\lambda = 0.01$), $\tau_{\text{base}} = 0.5$, $\alpha = 1.0$, arm-chain mask only.

C Architecture Details

Table 4 summarises the key dimensions of each Stage-2 component.

Fusion. For $r \in \{h, u, l\}$: $Z_r = (1-\Psi)\hat{Z}_r^b + \Psi\hat{Z}_r^s$, quantised as $\hat{c}_r = \arg\min_k \|e_k - Z_r\|_2$. The face region uses Z_e^b directly (no semantic stream). All regions decoded by frozen $\mathcal{D}^r$: $\hat{\mathbf{G}}^r = \mathcal{D}^r(\hat{c}_r)$.

Table 4: Stage-2 architecture summary. All transformer layers use $H=4$ heads, dropout 0.1, and batch-first layout. Stage-1 weights are frozen.

Component	**Details**	**Dims**
Inputs		
HuBERT encoder	2× Conv1d (k=3, p=1), BN, GELU	$1024 \rightarrow 256$
Onset/amplitude	2-layer MLP	$3 \rightarrow 256$
FastText	Linear	$300 \rightarrow 256$
Seed pose	VQEncoderV6 (3 layers, frozen)	$337 \rightarrow 768$
Speaker ID	Embedding table (25 entries)	$\rightarrow 768$
Shared backbone (both streams)		
Self-attention	TransformerEncoder, 1 layer, d_{ff}=1536	$d = 768$
Audio cross-attention	TransformerDecoder, 8 layers, d_{ff}=1536	$d = 768$
MGSC		
TM projections [Petrovich et al., 2023]	Linear (e_m, e_ε)	$256 \rightarrow 256$
Softmax memory gate	2× Linear	$256 \rightarrow 1$
Semantic cross-attention	TransformerDecoder, 1 layer, d_{ff}=512	$d = 256$
Output MLP	2-layer	$256 \rightarrow 256$
S-VIB		
Timing projection	Linear + LN + GELU	$256 \rightarrow 64$
Bottleneck heads	2× Linear (mean, log-var)	$320 \rightarrow 16$
Latent Sparsity Interpreter	2-layer MLP + softmax	$16 \rightarrow 2$
Region decoders (beat & semantic, per region $r \in \{h, u, l\}$)		
Region MLP	2-layer	$768 \rightarrow 256$
Temporal downsample	Conv1d, k=2, s=2	$L \rightarrow L/2$
HCA (3 decoders)	TransformerDecoder, 1 layer, d_{ff}=512	$d = 256$
Face (beat only)		
Face decoder	TransformerDecoder, 4 layers, d_{ff}=1536	$d = 768$
Downsample	Linear + 2× Conv1d (k=2,s=2) + MLP	$768 \rightarrow 256$
Codebook prediction heads (per region, 5 levels)		
Autoregressive decoder	TransformerDecoder, 3 layers, $H=8$, d_{ff}=1024	$d = 256$
Level classifier	2-layer MLP	$256 \rightarrow 256$
Total Stage-2 params		$\approx$ 122M

D User Studies Interface and Details

Figure 7 presents the evaluation interface used in our user study, which was developed in Qualtrics. During the evaluation, participants were required to first watch a co-speech gesture animation video and subsequently assess the quality of the generated gestures using a five-point Likert scale. The evaluation focused on three aspects: *Naturalness*, *Diversity*, and *Alignment with Speech Content and Timing*. In total, the questionnaire consisted of 24 video samples for evaluation. To minimise potential ordering effects and subjective bias, all videos were displayed in a randomised order for each participant. Participant recruitment and study administration were conducted through the Prolific platform, targeting participants from English-speaking countries.

To further ensure the reliability and validity of the collected responses, we incorporated additional attention-check questions throughout the evaluation process, as illustrated in Figure 8. In these questions, participants were asked to identify the topic of the narrated speech from several candidate options, such as *career*, *helping a friend*, or *internet*. Responses from participants who failed the attention-check questions were excluded from the final analysis. These quality-control measures were introduced to verify that participants carefully watched the videos and adequately understood the accompanying speech content during the evaluation.

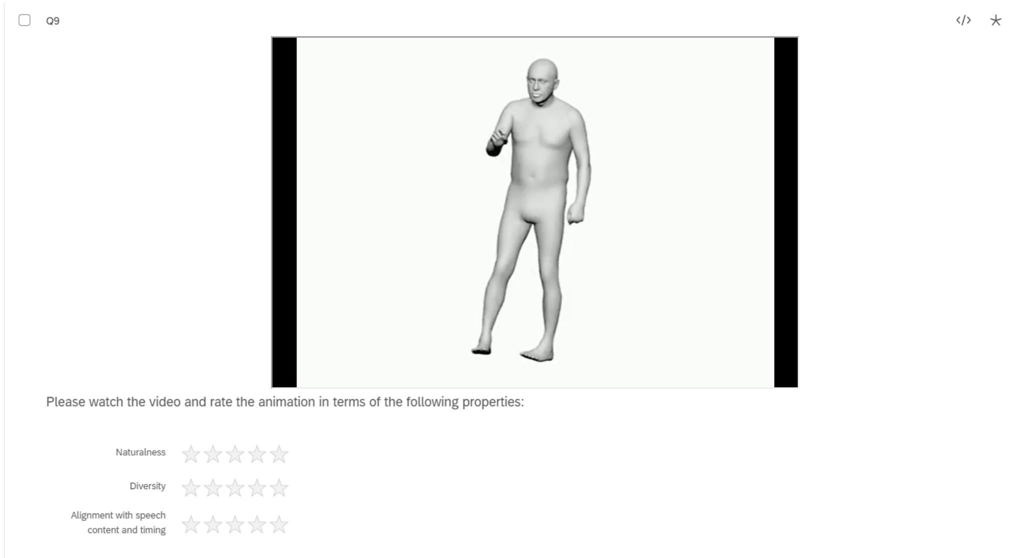

Figure 7: User study interface developed using Qualtrics. Participants were instructed to watch a co-speech gesture animation video and to evaluate the generated motions using a five-point Likert scale across three aspects: naturalness, diversity, and alignment with speech content and timing.

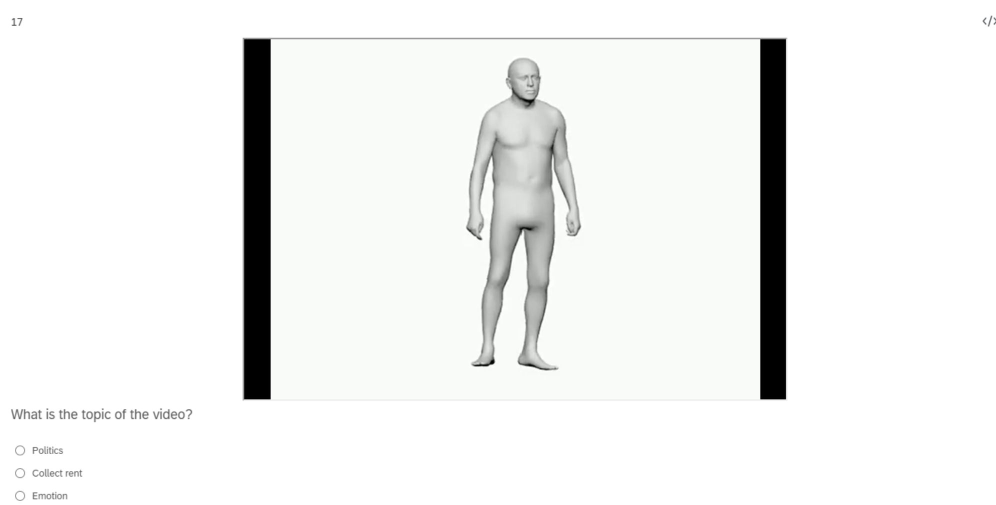

Figure 8: Attention-check interface employed during the user study to verify participant engagement and comprehension of the narrated content.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction accurately describe the paper's main contributions and experimental scope. The paper proposes DuoGesture, a dual-stream co-speech gesture generator that separates semantic and beat motion, introduces MGSC for motion-grounded semantic conditioning, uses S-VIB for stochastic frame-level stream selection, and applies IBP as an anthropometry-weighted inertial regulariser for beat motion. Quantitative results on BEAT2, component-wise ablations, qualitative comparisons, and a perceptual user study support these claims.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Sec. 5 discusses the main limitations: lack of evaluation across cultures and languages, dependence on pretrained text-to-motion representations, the simplified arm-chain biomechanical prior, and future directions for broader evaluation and improved metrics

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